

5.4. Doing Laundry Challenge

The robot assists with laundry by retrieving clothes from a washing machine and folding them neatly on a table. This test evaluates deformable object manipulation, appliance interaction, and task sequencing.

Main goal: Transport clothes to a table, and fold them.

Optional goals:

1. Opening the washing machine door
2. Retrieving clothes from inside the washing machine
3. Using a laundry basket for transportation
4. Folding multiple pieces of clothing and stacking them

Focus

Deformable object manipulation, bimanual grasping.

Setup

- **Locations:**

- **Start location:** Before the test, the robot waits outside the *Arena* and navigates to the laundry area when the door is open.
- **Test locations:** The testing area contains a washing machine, a folding surface nearby, and a laundry basket.

- **People:**

- No people are involved in the test, unless the robot requires human assistance.

- **Furniture:**

- **Laundry basket:** A lightweight basket requiring two arms to carry, placed near the washing machine. The basket already has 6–8 pieces of clothing placed inside.
- **Washing machine:** Positioned with its door closed at the start. Clothes are placed inside. There are 2–4 pieces of clothing in the washing machine.
- **Table:** Positioned near the washing machine. Used as the folding surface. There is a single piece of clothing loosely placed (or bunched up) on the table.

- **Objects:**

- **Clothes:** A set of 8–12 pieces of laundry made from cloth or other deformable fabric.

Procedure

1. **Location announcement:** At least two hours before the test, referees announce which washing machine and table will be used, along with the approximate location of each.
2. **Test start:** The robot enters the testing area once the arena door is opened.
3. **Retrieving laundry:** The robot retrieves the clothes from the laundry basket or directly from the washing machine and places them on the table.
4. **Folding:** The robot attempts to fold at least one piece of clothing neatly.

Additional rules and remarks

1. **Laundry Types:** The laundry will consist exclusively of T-shirts.
2. **Cloth placement:** Clothes must be placed on the table before folding. Folding on the floor is not allowed.
3. **Folding quality:** Folding is evaluated by neatness and whether the cloth is flattened and stacked (see Appendix A for examples. Scores are assigned holistically based on the final folding quality after applying any penalties).
4. **Multiple items:** Additional points are awarded for folding multiple pieces of clothing and stacking them.
5. **Basket transportation:** Using the basket to transport clothing from the washing machine to the table. The basket must be placed in reach of the folding surface.
6. **Human Assistance:** Scores are reduced if human assistance is received, in particular for:
 - opening the washing machine door
 - handing clothes to the robot
 - **Folding:** Penalties are proportional to the amount of help provided:
 - Minimal help (e.g., smoothing wrinkles, stabilizing cloth): small penalty (100 points).
 - Partial folding (e.g., folding one half or aligning edges): moderate penalty (200 points).
 - Major help (e.g., completing entire fold/stack): maximum penalty.

OC Instructions

At least two hours before the test:

- Announce which washing machine and table will be used.

Referee Instructions

The referee needs to:

- Ensure the washing machine and basket contains the laundry before the test begins.
- Close the washing machine door at the start.
- Verify that the folding table is clear before the test.

Score sheet

The maximum time for this test is **7:00** minutes.

Action	Score
Main Goal	
Navigating to the laundry area	15
Picking up a piece of clothing from the basket	100
Picking up multiple at once	-100
Folding a piece of clothing	800
Quality penalties	-800
Human assistance: Flattening/Arranging clothing before folding	-200
Human assistance during folding will apply a percentage penalty according to the amount of help	-800
Extra Rewards	
Opening the washing machine door	300
Removing one or more pieces of clothing from the washing machine	300
The clothing touches the floor	-200
Using the basket for transportation	300
Laundry is dropped or otherwise lost during transportation.	-200
Folding additional clothes (per item)	5×400
Quality penalties	-400
Human assistance: Flattening/Arranging clothing before folding	5×-100
Human assistance during folding will apply a percentage penalty according to the amount of help	5×-400
Stacking folded clothes neatly (per item)	6×100
Penalties	
Human assistance: environment changes (per item)	-40
Special Penalties & Bonuses	
Not attending (see sec. 3.8.1)	-500
Using alternative start signal (see sec. 3.6.8)	-100
Outstanding performance (see sec. 3.8.3)	441
Total Score (excluding special penalties & standard bonuses)	4415

Appendix A

Doing Laundry: Folding Score Examples

Definition

Folding a t-shirt means taking a flat, spread-out shirt and folding one part over another along an intended fold line, reducing the shirt size in at least one dimension.

A fold is considered successful if:

- the folded section lies flat without significant bunching,
- edges are reasonably aligned, and
- no major unintended creases or folds appear outside the intended fold line.

A folding task consists of a sequence of folds resulting in a neat, approximately rectangular shape.

Folding Quality Evaluation

Folding quality is evaluated based on neatness, flatness, and alignment

All percentages in this appendix refer to the maximum available folding score. Assistance penalties are applied as deductions from the maximum folding score, independent of the achieved folding quality score.

Penalties for human assistance during folding are proportional to the amount of help provided:

- **Minimal help** (e.g., smoothing wrinkles, stabilizing cloth): small penalty (12.5% points).
- **Partial folding** (e.g., folding one half or aligning edges): moderate penalty (25% points).
- **Major help** (e.g., completing entire fold/stack): maximum penalty (100% points).

The penalties are indicative examples and referees may adjust penalties proportionally to the level of assistance provided.

Examples



(a) (failure **0 points**): The shirt is completely unfolded; no fold was attempted.



(b) (laying flat **25%**): A human spread the clothing flat (**25%** point penalty).



(c) (initial folding **31.25%**): The robot only grasped one side, a human assisted, folded one half of the shirt (**25%** point penalty).

Figure A.1.: Example of different foldings and corresponding rulings



(a) (partial success **37.5%**) The shirt is folded but shows noticeable misalignment or slight bunching; neater than Figure A.1b.



(b) (moderate success **50%**) The shirt is folded using a basic technique; the result is acceptable but not neat.



(c) (good **75%**): The shirt is well folded with only minor neatness issues; edges are mostly aligned.

Figure A.2.: Example of different foldings and corresponding rulings



Figure A.3.: (partial success **87.5%**): The shirt appears properly folded from the front, but the back shows unintended folds where fabric bends over itself outside the intended fold lines.



Figure A.4.: (success **Full points**): The shirt is neatly folded, flat, and well aligned.

